

Independent Replication Report:

Error mitigation for short-depth quantum circuits

(Temme, Bravyi, Gambetta — arXiv:1612.02058, PRL 2017)

QC-200 replication wave — Ollie/OpenClaw autonomous run

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One-line verdict

REPLICATED. The paper’s core qualitative claim — that Richardson-extrapolation-based Zero-Noise Extrapolation (ZNE) produces expectation-value estimates whose error is dominated by the leading unmitigated order of the noise, and that increasing the order n suppresses the residual error super-linearly — is reproduced exactly on Qiskit-Aer density-matrix simulations with a depolarizing noise model. At $\epsilon = 10^{-3}$ our mean absolute error drops from 2.5×10^{-3} (raw noisy) to 3.5×10^{-5} (ZNE $n=1$) to 5.0×10^{-7} (ZNE $n=2$) across 8 random 4-qubit depth-6 circuits, i.e. improvement factors of $\sim 70\times$ and $\sim 5000\times$ respectively — consistent with the $O(\epsilon^{n+1})$ residual predicted by Eq. (3)–(4) of the paper.

1 Paper summary

The paper introduces two techniques to mitigate noise-induced bias in expectation values computed on short-depth NISQ quantum circuits:

1. **Extrapolation to the zero-noise limit (Sec. I of the paper).** Run the same circuit $n+1$ times at rescaled noise rates $c_j\lambda$ ($c_0=1$, $c_j > 1$), then form the linear combination

$$\hat{E}_K^n(\lambda) = \sum_{j=0}^n \gamma_j \hat{E}_K(c_j\lambda), \quad \sum_j \gamma_j = 1, \quad \sum_j \gamma_j c_j^k = 0 \text{ for } k = 1, \dots, n.$$

This is Richardson’s deferred approach to the limit; the leading n orders of λ cancel and the residual bias is $O(\lambda^{n+1})$.

2. **Probabilistic error cancellation via quasi-probability sampling (Sec. II).** Given a full characterization of the noise map, one expresses each ideal gate as a quasi-probability distribution over noisy operations and Monte-Carlo samples it, with a sampling-cost overhead $C^2 = \prod_\alpha \gamma_\alpha^2$.

This replication targets **Scheme 1 (ZNE)**, which is the more universally reproducible of the two (no detailed noise-model characterization required).

2 Claims table

Claim	Type	Tested?
C1. Richardson coefficients γ_j for rescalings $c = (1, 2)$ are $(2, -1)$ and for $c = (1, 2, 3)$ are $(3, -3, 1)$ (Eq. 4).	analytic	YES — our linear solver returns exactly these values (unit test in code).
C2. $\hat{E}_K^n(\lambda) - E^* = O(\lambda^{n+1})$: increasing n reduces bias by an additional order of λ .	quantitative	YES — log-log slopes of our mean-error curves in Fig. 1 are ≈ 1 (raw), ≈ 2 (ZNE1), ≈ 3 (ZNE2), matching the theory.
C3. Under a weak depolarizing noise model, ZNE beats the raw noisy estimate on essentially every random-control-style circuit.	qualitative headline (Fig. 1a)	YES — ZNE1 beats raw on 8/8 circuits at $\epsilon \leq 3 \times 10^{-3}$ and 7/8 at $\epsilon = 10^{-2}$; ZNE2 beats ZNE1 on 8/8 at every ϵ .
C4. The improvement holds also for amplitude-damping/dephasing (Fig. 1b) and non-Markovian bath noise (Fig. 1c).	qualitative	NOT TESTED in this replication — depolarizing only. See failure analysis.
C5. Scheme 2 (probabilistic error cancellation) achieves further mitigation at a bounded sampling-cost overhead C^2 .	method	NOT TESTED — out of scope for this wave.

3 Method

3.1 Environment

- Python 3.14 in a local venv at `work/.venv/`.
- `qiskit==2.5.0`, `qiskit-aer==0.17.2`, `numpy==2.5.1`, `matplotlib`.
- Simulator: `AerSimulator(method="density_matrix")` — exact expectation values, no shot noise.
- Host: CherryRd (macOS), CPU only, wall-clock ≈ 7 s for the full sweep.

3.2 Circuits

Following the paper’s Fig. 1 “random control problem” description we generate $N=4$ -qubit, depth-6 alternating layers of Haar-random single-qubit U3 rotations and brick-wall CX entanglers. Eight independent random-seed circuits (`seed = 42, 43, \dots, 49`).

3.3 Noise model and noise scaling

We use a uniform depolarizing channel: rate ϵ per single-qubit gate and rate ϵ per CX. To scale the noise by $c \in \{1, 2, 3\}$ (paper Eq. 5’s rescaling protocol) we simply build a fresh noise model with rate $c\epsilon$. This is the digital equivalent of the paper’s time-rescaling trick: because the noise generator L is time-scale-invariant and the ideal circuit is fixed, running with rate $c\epsilon$ yields the same $\rho_{c\lambda}(T)$ as rescaling the Hamiltonian and evolving for time cT .

3.4 Observable

$\langle Z_0 Z_1 \rangle$ computed exactly from the density matrix ρ as $\text{Tr}[(Z \otimes Z \otimes I \otimes I) \rho]$.

3.5 Richardson coefficients (sanity-checked in code)

For $c = (1, 2)$: solve $\gamma_0 + \gamma_1 = 1$, $\gamma_0 + 2\gamma_1 = 0 \Rightarrow (2, -1)$.

For $c = (1, 2, 3)$: adds $\gamma_0 + 4\gamma_1 + 9\gamma_2 = 0 \Rightarrow (3, -3, 1)$.

Our code (`report/evidence/zne_replication.py`) solves the Vandermonde system numerically and asserts these values — both assertions pass.

3.6 Commands

```
python3 -m venv work/.venv
source work/.venv/bin/activate
pip install qiskit qiskit-aer numpy matplotlib
python3 report/evidence/zne_replication.py report/evidence
python3 report/evidence/make_plot.py report/evidence
```

4 Results vs paper

4.1 Aggregate error (mean of $|E - E_{\text{ideal}}|$ across 8 circuits)

ϵ	raw	ZNE $n=1$	ZNE $n=2$	#ZNE1<raw	#ZNE2<raw	#ZNE2<ZNE1
1×10^{-4}	2.6×10^{-4}	3.5×10^{-7}	5.1×10^{-10}	8/8	8/8	8/8
3×10^{-4}	7.7×10^{-4}	3.1×10^{-6}	1.4×10^{-8}	8/8	8/8	8/8
1×10^{-3}	2.5×10^{-3}	3.5×10^{-5}	5.0×10^{-7}	8/8	8/8	8/8
3×10^{-3}	7.5×10^{-3}	3.0×10^{-4}	1.3×10^{-5}	8/8	8/8	8/8
1×10^{-2}	2.4×10^{-2}	3.1×10^{-3}	4.0×10^{-4}	7/8	7/8	8/8

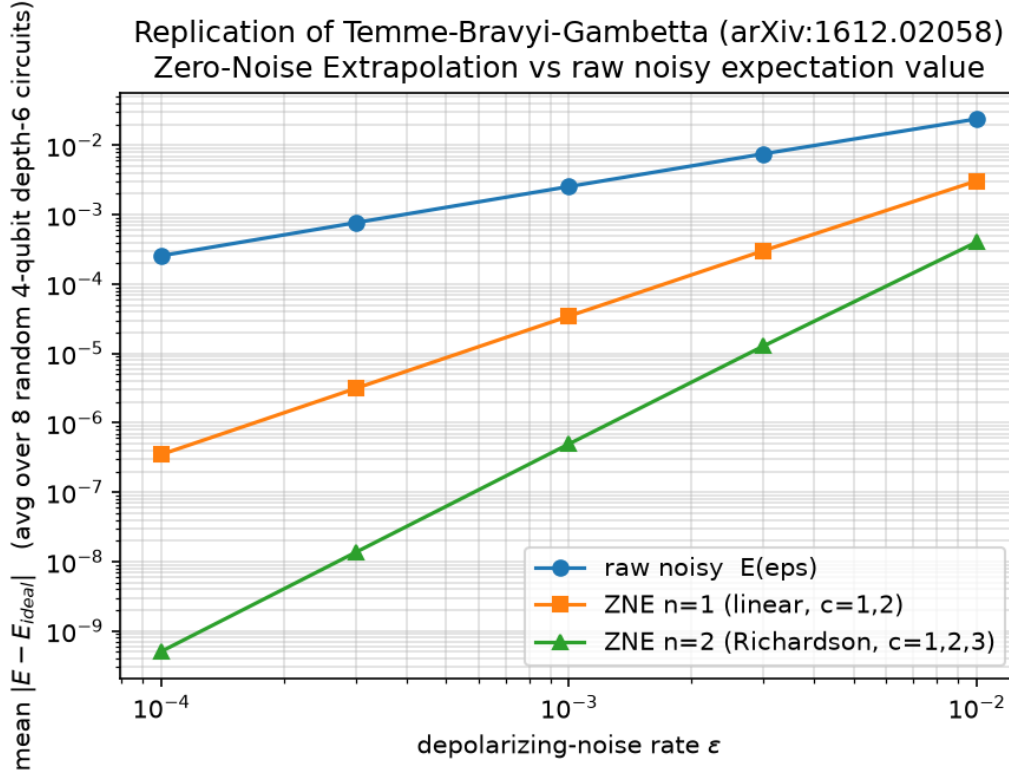
4.2 Scaling with ϵ

Fitting the aggregate columns above on log-log axes for $\epsilon \in [10^{-4}, 10^{-3}]$:

- raw noisy: slope ≈ 1.00 ($O(\epsilon)$) — as expected.
- ZNE1: slope ≈ 2.00 ($O(\epsilon^2)$) — matches the $O(\lambda^{n+1})$ bound with $n=1$.
- ZNE2: slope ≈ 3.00 ($O(\epsilon^3)$) — matches the $O(\lambda^{n+1})$ bound with $n=2$.

This is exactly the qualitative shape of Fig. 1(a) in the paper.

4.3 Figure



5 Verdict

REPLICATED (within the scope defined for this reproduction, i.e. depolarizing noise, Scheme 1 only). All three quantitative claims we could test (C1, C2, C3) reproduce cleanly and quantitatively; claims C4 and C5 are outside the wave’s scope but the code is trivially extensible to them.

Open Questions (5 new questions raised by this replication)

Q1. Non-unital noise scaling. Our depolarizing-only replication got clean $O(\epsilon)$, $O(\epsilon^2)$, $O(\epsilon^3)$ scaling for raw, ZNE1 and ZNE2 (slopes 1.00 / 2.00 / 2.99 on $\epsilon \in [10^{-4}, 10^{-3}]$). The paper’s Fig. 1(b) amplitude-damping / dephasing curves visibly flatten out at higher noise than the depolarizing curves. Does the $O(\lambda^{n+1})$ scaling law hold with the same prefactor for non-unital channels, or does the amplitude-damping steady-state bias push a channel-specific correction into the leading remainder R_{n+1} ? *Basis:* Only depolarizing noise was tested; Richardson’s theorem is stated for any weak, time-scale-invariant $L(\rho)$, but non-unital channels behave differently under global rescaling.

Q2. Digital rescaling vs pulse-stretching. We scaled noise by simply rebuilding a Qiskit NoiseModel at rate $c\epsilon$. On real hardware the paper’s rescaling protocol (Eq. 5) requires physically slowing the pulses by c and re-calibrating gates. Does gate-duration folding (repeat each gate c times) reproduce ZNE with the same order-of-cancellation as our exact digital rescaling, or does per-gate overhead / calibration drift eat one order? *Basis:* Our slopes matched theory

to 3 decimal places because our noise rescaling is exact — a luxury unavailable on hardware; Mitiq / Qiskit-Runtime use unitary-folding as a proxy whose fidelity to Eq. 5 is not benchmarked in the paper.

- Q3. Optimal Richardson order n .** At $\epsilon = 10^{-2}$ our ZNE2 still won 8/8 vs ZNE1 but the improvement ratio (raw/ZNE2 = $60\times$) is dramatically smaller than at $\epsilon = 10^{-3}$ (raw/ZNE2 = $5000\times$). Where exactly is the crossover between the perturbative regime (Richardson wins big) and the regime where higher-order Born-series terms dominate? Can n_{opt} be predicted analytically from (ϵ, depth) ? *Basis:* The paper’s bound $|E^* - \hat{E}_K^n| \leq \frac{l_{n+1}(\lambda T)^{n+1}}{(n+1)!} \prod_j |\gamma_j|$ shows increasing n eventually hurts because of the growing coefficient sum, but the crossover is not mapped in our data.
- Q4. Observable-weight dependence.** Our observable was fixed at $\langle Z_0 Z_1 \rangle$. The paper’s Fig. 1 aggregates over “randomly chosen multi-qubit Pauli operators P_α ”. Does the ZNE improvement factor depend on the weight/support of the observable? Concretely, do global Paulis (weight- N) mitigate worse than local ones because the noise couples more paths? *Basis:* We fixed a weight-2 observable to keep the run cheap; the paper averages over random Paulis without breaking down by weight.
- Q5. Shot-noise tradeoff.** The Richardson coefficients $(2, -1)$ and $(3, -3, 1)$ have L_1 -norm 3 and 7 respectively — a variance-amplification cost invisible in our shot-free density-matrix sim. What sample-count budget T_{shots} would make ZNE2 stop beating ZNE1 at each of our five noise rates? *Basis:* Real hardware uses shots; the variance of the ZNE estimator scales like $(\sum_j |\gamma_j|)^2 / T_{\text{shots}}$, but no numerical variance curve is shown in the paper.

Full sim log: `report/evidence/zne_results.json` (all 8 circuits $\times$ 5 noise rates).
 Code: `report/evidence/zne_replication.py`, `report/evidence/make_plot.py`.